

A1

a) $8.26; 18.3 \text{ dB}$

b) $25, 28 \text{ dB}$

c) $0.83, -1.66 \text{ dB}$

A2

$$A_s = 83.3 = 38.4 \text{ dB}$$

$$A_p = 13.9 \cdot 10^6 = 71.4 \text{ dB}$$

$$v_{i_{\text{rms max}}} = 84 \text{ mV}$$

A3

744 or $57.4 \text{ dB}; 412$ or 52.3

A4

$S-A-B-L$

A5

...

1.56 or 38.1 V/V

A6

A7

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$$A_v = \frac{G R_2}{1 + G R_2}$$

$$R_i = R_1$$

$$R_o = \frac{R_2}{1 + G R_2}$$

Ⓑ

$$A_v = \frac{-\beta R_3}{R_1(1 + \beta R_3) + R_2}$$

$$R_i = R_1(1 + \beta R_3) + R_2$$

$$R_o = \frac{\beta R_3(R_1 + R_2)}{\beta R_3(1 + \beta R_3) + R_2}$$

A8

$$A_8 = \frac{2 R_8 (6 R_3 R_4 - \kappa R_1)}{R_5 (R_2 + R_4 + R_8) (R_1 + R_5 + R_6 + R_7)}$$

$$R_i = R_s ; \quad R_o = \frac{R_8 (R_2 + R_4)}{R_2 + R_4 + R_8}$$